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**Operating Instructions
for
PD-471
PMT Housing
With HV Supply**

Model PD-471

PMT Detector Housing with Integral High Voltage Power Supply

Description:

The PD-471 PMT Detector Housing is designed for use with Acton Series Monochromators and the SpectraSense /NCL Data Acquisition System. The PD-471 consists of the PMT detector housing for use with standard 1.125 inch diameter 9 stage side window PMTs, a mounting flange for attachment to Acton Series Monochromators, a control cable for attachment to the NCL Data Acquisition Module to allow control of high voltage and a signal cable for attaching to the signal input of the NCL. If purchased with a PMT from Princeton Instruments-Acton, the PMT is normally installed in the PD-471 and is shipped as a complete detector assembly.

The detector assembly provided with your order is as follows:

PD-471 Serial Number: _____

PMT Provided with Your Order: ____ P1 (1P28) ____ P2 (R928) ____ P3 (R5108)
Other PMT _____

Mounting Instructions:

The PD-471 mounts directly to the exit slit of the Acton Series Monochromators. The detector should be positioned with the BNC signal connector and 6 pin control connector pointing up (refer to the manufacturer's data sheet attached). The PD-471 is mounted directly against the exit slit such that the four mounting holes of the mounting flange align correctly with the tapped mounting holes of the exit slit. Insert the four 8-32 mounting screws and tighten. The detector should now be mounted correctly.

Connect the control cable provided with the PD-471 from the 6 pin connector on the PD-471 to the channel 1, 2 or 3 control input on the NCL. Connect the signal cable provided with the PD-471 from the BNC connector on the PD-471 to the corresponding channel 1, 2 or 3 input on the NCL. The PD-471 is now ready for use.

Note: If the PD-471 is to be used with the FA-448 filter wheel at the exit slit, the mounting procedure requires three (3) 8-32 screws 1.375 inches long. Insert the three screws into the detector flange, through the filter wheel, and into the tapped holes of the slit or adapter plate. Tighten the screws to secure the accessories. The FA-448 Filter Wheel is more often attached to the entrance slit, however.

